FLAT FINITE TYPE RING MAPS AND PROPER FLAT MORPHISMS WITH CONTROLLED FIBRES THAT ARE NOT OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The four context paragraphs map one-to-one onto the four enumerated items; the (R, I) base data is the delivered L48315 example, making the chapter self-contained.

Editorial note. The label lemma-finite-type-flat-not-finitely-presented names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3050-3059: R is a ring with an ideal I such that R/I is a finite flat module but not projective (explicit example delivered as card L48315, section-finite-flat-not-projective); this means I is not finitely generated (Algebra, finitely-generated-pure-ideal) and I = I^2 (Algebra, pure); the base scheme is S = Spec(R)
- Context span 3062-3065: the ring map R -> R + (R/I)epsilon with epsilon^2 = 0 is finite, flat, not of finite presentation, with all fibre rings complete intersections and geometrically irreducible
- Context span 3068-3073: A = R[x, y]/(xy, ay for a in I) is as an R-module the direct sum of R y^i (i >= 0) and (R/I) x^j (j > 0); R -> A is flat finite type not of finite presentation, each fibre ring isomorphic to kappa(p)[x, y]/(xy) or kappa(p)[x]
- Context span 3076-3082: B = R[X_0, X_1, X_2]/(X_1 X_2, a X_2 for a in I) gives Proj(B) -> S proper flat not of finite presentation with fibres P^1_s or the vanishing locus of X_1 X_2 in P^2_s (a projective nodal curve of arithmetic genus 0)
- Context span 3085-3090: M = R + R + R/I, B = Sym_R(M), X = Proj(B): X -> S is proper flat not of finite presentation with geometric fibres P^1_s or P^2_s, in particular smooth and geometrically irreducible

Source reference. Original lemma 05KK: sources/stacks-chow/tags.txt, source lines 7204–7204. Prerequisite pointer invoked at 3118; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04PS: sources/stacks-chow/tags.txt, source lines 6125–6125. Prerequisite pointer invoked at 3120; it names a standard fact of a

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sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 052H: sources/stacks-examples/pinned-original-source.tex, source lines 2620–2620. Prerequisite pointer invoked at 3115; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let R be a ring which has an ideal I such that R/I is a finite flat module but not projective, see Section 052H for an explicit example. Note that this means that I is not finitely generated, see Algebra, Lemma 05KK. Note that $I = I^2$, see Algebra, Lemma 04PS. The base ring in our examples will be R and correspondingly the base scheme $S = \mathrm{Spec}(R)$.

Consider the ring map $R \to R \oplus R/I\epsilon$ where $\epsilon^2 = 0$ by convention. This is a finite, flat ring map which is not of finite presentation. All the fibre rings are complete intersections and geometrically irreducible.

Let $A = R[x, y]/(xy, ay; a \in I)$. Note that as an R-module we have $A = \bigoplus_{i \geq 0} Ry^i \oplus \bigoplus_{j>0} R/Ix^j$. Hence $R \to A$ is a flat finite type ring map which is not of finite presentation. Each fibre ring is isomorphic to either $\kappa(\mathfrak{p})[x, y]/(xy)$ or $\kappa(\mathfrak{p})[x]$.

We can turn the previous example into a projective morphism by taking $B = R[X_0, X_1, X_2]/(X_1X_2, aX_2; a \in I)$. In this case $X = \mathrm{Proj}(B) \to S$ is a proper flat morphism which is not of finite presentation such that for each $s \in S$ the fibre X_s is isomorphic either to $\mathbf{P}^1_s$ or to the closed subscheme of $\mathbf{P}^2_s$ defined by the vanishing of X_1X_2 (this is a projective nodal curve of arithmetic genus 0).

Let $M = R \oplus R \oplus R/I$. Set $B = \mathrm{Sym}_R(M)$ the symmetric algebra on M. Set $X = \mathrm{Proj}(B)$. Then $X \to S$ is a proper flat morphism, not of finite presentation such that for $s \in S$ the geometric fibre is isomorphic to either $\mathbf{P}^1_s$ or $\mathbf{P}^2_s$. In particular these fibres are smooth and geometrically irreducible.

Lemma 0.1. *There exist examples of*

1. *a flat finite type ring map with geometrically irreducible complete intersection fibre rings which is not of finite presentation,*
2. *a flat finite type ring map with geometrically connected, geometrically reduced, dimension 1, complete intersection fibre rings which is not of finite presentation,*
3. *a proper flat morphism of schemes $X \to S$ each of whose fibres is isomorphic to either $\mathbf{P}^1_s$ or to the vanishing locus of X_1X_2 in $\mathbf{P}^2_s$ which is not of finite presentation, and*
4. *a proper flat morphism of schemes $X \to S$ each of whose fibres is isomorphic to either $\mathbf{P}^1_s$ or $\mathbf{P}^2_s$ which is not of finite presentation.*

Proof. See discussion above. □